Finding the survivors who can't call for help

Rubble search still waits for a knock or a shout. Larpverse sends heavy drones that map the pile, call out, and land on the rubble and scan it with a breathing-detection radar, so unconscious and silent people get on the dig list in the first hour.

[S] sourced [D] derived from sources [A] our assumption · simulation results are marked as simulation

72h The problem

≥28%

of injured Bam 2003 survivors were **unconscious when rescued**. Listening devices and call-outs can't hear them. Survivors-only sample, so the true share among the trapped is likely higher.

[S] Nia, Nafissi & Moharamzad 2008, Prehosp Disaster Med 23:263 · via Rom & Kelman 2020

164 → 7 → 4 → 0

people rescued alive at Weiguan Jinlong (Tainan 2016) on days 1, 2 and 3. **Nobody alive after 62.3 h.** 115 of 309 residents died; about 2,000 rescuers and 23 dogs worked the site.

[S] Chiu & Tseng 2018 · Tainan City Fire Bureau · Pan et al. 2019, J Formos Med Assoc

<0.85%

International rescue teams' live rescues as a share of deaths, in all 14 earthquakes reviewed. The first team arrived **12–56 h** after the quake, at about **US$0.8–0.94M per life saved**.

[S] Rom & Kelman 2020, BMJ Global Health 5:e002398

~300 of 53,537

Turkey 2023: about 5,000 international rescuers in 90 teams saved about 300 lives while 53,537 died. There were 26,000 buildings to search. Crews switched off generators and asked victims to scratch with a fingernail.

[S] INSARAG after-action review · AP · NTV

77.5%

of the 35,000 people dug out of rubble in Kobe 1995 were rescued by neighbours and family, not professionals. **Speed and local capability decide outcomes.**

[S] Kobe City / ADRC

0

documented cases of a drone **locating a living person inside a collapse** in 15 major collapses reviewed. Drones have been used for overview, mapping and lighting only.

[D] review of 15 collapses (docs/CASE_STUDIES.md)

T Survival falls by the day

SHARE OF PEOPLE BROUGHT OUT ALIVE	DAY 1	DAY 2	DAY 3	AFTER 72 H
Kobe 1995 (fire brigade rescues) [S]	80.5%	28.5%	21.8%	~6%
Weiguan Jinlong 2016 [D]	92.7%	25.9%	~57% (n≈7, weak)	0 alive after 62.3 h

The pattern repeats: Surfside 2021, where a woman's voice was heard on day 1 but she couldn't be reached; Rana Plaza 2013, where a woman found alive after 100 h died when a fire ended 20 h of tunnelling; Hualien 2018, where the search stopped at 106 h with the last victims under a load-bearing beam. People who can be heard get found. People who can't, often don't.

L? What Larpverse does

The loop, fully autonomous

- **Recon:** three heavy-lift drones map the ruin with 128-channel LiDAR. Satellite imagery is stale after a quake.
- **Listen and call out:** drones perch with rotors off. A loudspeaker asks "if you can hear me, knock three times" in Mandarin, Taiwanese Hokkien and English.
- **Land and scan:** each drone lands on a flat spot on the pile over a lead, and its downward-facing radar reads breathing through the rubble. It gives a condition hint, such as "irregular breathing with pauses — check first".
- **Rescue Wi-Fi:** every drone serves an open hotspot. A trapped person's phone can tap "I'm trapped" in six languages.
- **AI decides, code verifies:** an agent ranks the dig order. Rules block unsafe or overconfident wording, and a deterministic fallback runs if the AI is unavailable.

Why a drone-carried radar is the gap

Breathing radar already works on rubble when a person carries it there. **NASA/DHS FINDER** found two heartbeats under each of two collapsed buildings in Nepal 2015, and 4 men were dug out. It's lab-rated through **9 m of rubble or 6 m of solid concrete**. GSSI LifeLocator units (about 270 MHz) were used at Sichuan 2008 and Haiti 2010.

Today a rescuer carries these units onto an unstable pile, one spot at a time. Larpverse **gives that class of radar wings and an autonomous planner**: each drone carries a FINDER- or LifeLocator-class sensor and lands it on the pile.

[S] DHS 2015 · NASA JPL · GSSI · Frontiers in Earth Science 2023

+ Measured so far

53% in 30 min
of trapped people inside a drone-flagged search sector within 30 minutes of launch (median 8 min), and 83% within an hour, on a Weiguan-scale site.
[D] simulation, 10 missions

~49 min
from the quake to the first radar ping in a Weiguan-like Tainan scenario (range 23–94). National special teams took up to 12 h at Weiguan.
[D]/[A] docs/DEPLOYMENT.md

0 false pings
across every evaluation run. Radar returns from debris, animals, machinery and rotors are rejected (90% of non-human signatures).
[D] simulation; signature library

6 / 6
real breath-holds detected with no false alarms. Breathing rate within about 0.5 per minute of a clinical sensor on real radar recordings.
[S] Schellenberger 2020 open dataset (line of sight)

86%
of breathing conditions identified correctly in synthetic 60 s scans. Rapid shallow breathing (possible distress) is caught 100% of the time.
[D] simulation, docs/VITALS.md

3×
more survivors pinged than nearest-pile or lawnmower search with the same drones and budget (17/60 vs 6/60 and 5/60). This is an upper bound.
[D] 20-scenario simulation, docs/EVAL_GRID.md

RADAR DETECTION DEPTH	FIGURE	BASIS
Our simulation, deliberately conservative	2.9 m beneath the landed drone (45 s scan)	[D] literature concrete loss (10–20 dB/m), realistic chest motion
Commercial rubble life detector (LifeLocator)	4.5–7 m breathing	[S] vendor and trade press
NASA/DHS FINDER lab rating	9 m rubble / 6 m solid concrete	[S] NASA JPL

$ Economics

CITY KIT: 3 RADAR DRONES, VAN, TRAINING	CENTRAL	RANGE
Up front	$673k	$361k–1.10M
Per year to run	$193k	$70–469k
10-year total cost	$2.78M	$1.06–6.31M

LIVES PER DECADE TO PAY FOR ITSELF	CENTRAL
At the US DOT value of a life ($13.7M)	0.20
At Taiwan's benefit-transfer value ($5M)	0.56
At international USAR spend per life saved ($0.93M)	2.99

A kit costs about the yearly price of a police first-responder drone contract ($140–435k) or 8.5 dog teams over ten years. Public-safety drones were a **$2.6B market in 2024, projected to reach $5.8B by 2030**. Taiwan is the first market, because government drone tenders already exclude PRC parts, which favours our non-DJI build. [S]/[D]/[A] 87 tagged parameters, docs/ECONOMICS.md

TW Deploying it in Taiwan

99%

chance of an M≥6.5 quake somewhere in Taiwan within 50 years (54% for M≥7). About half of homes in Tainan and Kaohsiung, and 70% in Taipei, are over 30 years old.

[S] CWA figures via news report · housing statistics

47 kits

cover more than 90% of Taiwan's urban land within 20 minutes: **US$15.5M** to buy, about $2.3M a year to run. The kits ride on the existing fire service: 209 rescue-equipment trucks and 10 graded USAR teams.

[D] docs/DEPLOYMENT.md

250 min

of heavy-drone flight from half a car tank through a generator. A rescue truck with a 25 kW generator keeps about 4 drones working, so **no city hub is required**: retrofit onto existing trucks.

[D] DJI generator and battery specs; docs/IMPACT_EVIDENCE.md §6

72 h

of operation per kit on about 510 L of diesel, 3 battery sets per drone and a 7-person crew in two 12 h shifts.

[D] generator spec sheet; docs/DEPLOYMENT.md

+ Impact model (preliminary)

IF THE RADAR PERFORMS AS DESIGNED	CENTRAL	90% RANGE
Extra survivors among the 113 people brought out of Weiguan after day 1	~+6	0.6–14
Extra survivors per 100 trapped, 3×3 test site	+3.0	1.1–6.6
Rescuer-hours tied up before each person is found	−15%	

This is a model, not a claim about what happened. There's no time-of-death data for Weiguan, extrication remains the bottleneck, and the full-scale site is still being tuned. docs/IMPACT_MODEL.md

! What we claim, and what we don't

- A drone that lands a breathing radar on the pile fills a gap that no deployed system covers.
- It ranks where to dig sooner, including people who make no sound.
- Drone finds get the INSARAG "V" (victim suspected) mark for rescuers to confirm.
- Never "cleared": a searched zone without a ping is "searched, not cleared".
- No faster digging: extrication stays with rescue crews.
- The radar results are simulation; real validation so far is line-of-sight.

Key references: Rom A, Kelman I. Search without rescue? BMJ Glob Health 2020;5:e002398, doi:10.1136/bmjgh-2020-002398 · Nia MS, Nafissi N, Moharamzad Y. Prehosp Disaster Med 2008;23:263–8, doi:10.1017/S1049023X00064979 · Pan et al. 2019, J Formos Med Assoc · Chiu & Tseng 2018 · NASA JPL / DHS FINDER 2015 · INSARAG Türkiye after-action review. All other figures and their tags are in the repository docs named above.